Data Science 500 Assignment

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Qualification: BSc IT Year of Study: 1st Semester: 2nd

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: Date: 2026-10-22

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Title: Book Catalogue Pricing Analytics: An Exploratory and Predictive Analysis
Qualification: [BSc IT]
Student Name: [Nolwazi Rivoni Slindile Maswanganye]
Date: [2026-10-22]

1. Introduction

The purpose of this project was to analyse the pricing landscape of the Books to Scrape website and build a predictive model. As a Junior Data Scientist assisting a publishing analytics team, the goal was to extract the full catalogue of 1,000 books to understand what factors influence book prices across different genres and rating levels.

2. Methodology

- **Data Acquisition:** A Python script using requests and BeautifulSoup was used to scrape all 1,000 books from the website. We extracted the title, price, star rating, stock availability, stock quantity, category, and description.
- **Data Cleaning:** The raw data required cleaning to be usable. The price text (e.g., "£51.77") was converted into a float number. Star ratings (e.g., "Three") were mapped to integers (1-5). Stock availability was converted to binary (1 for in stock, 0 for out of stock), and descriptions were cleaned into word counts.
- **Data Visualization:** Exploratory Data Analysis (EDA) was conducted using Matplotlib to plot price distributions, average price by rating, and the most expensive categories.
- **Predictive Modelling:** A Linear Regression model was built using scikit-learn. The target variable was Price, and the features used were Rating, Stock Quantity, Availability, and Category (using one-hot encoding).

3. Key Insights from Visualizations

- **Price Distribution:** The histogram shows that most books are priced between £10 and £40, with a few expensive outliers reaching above £50.
- **Price by Rating:** The bar chart reveals that star rating does not heavily influence price. Books with 1-star ratings and 5-star ratings have very similar average prices (all hovering around £30–£35).
- **Top Categories:** The most expensive categories (such as "Suspense" and "Science") average around £50, but the majority of categories sit closely together in the £25–£35 range.

4. Predictive Modelling Results

The Linear Regression model yielded the following metrics:

- **R-squared:** -0.107
- **MAE:** £12.62

- **RMSE:** £15.10

Interpretation:
The model performed poorly. A negative R-squared indicates that the model's predictions are worse than simply guessing the average price of a book every time. The MAE shows that the model's predictions were off by an average of £12.62.

Model Limitations:
This failure is an important real-world finding. It proves that a book's price is not strongly correlated with its star rating, stock quantity, or genre. To accurately predict book prices, a dataset would need to include features we did not scrape, such as the publisher, page count, book format (hardcover vs. paperback), and publication year.

5. Recommendations
Based on the data, we recommend the following:

- **For Retailers:** Because rating and genre are poor indicators of price, retailers should not assume they can charge a premium purely based on a book's star rating. They must look at external market factors.
- **For Publishers:** Pricing strategies should focus on the physical attributes of the book (e.g., page count, hardcover editions) rather than the genre category, as the data shows genres are priced very similarly.
- **For Data Buyers:** If the analytics team wishes to build a better predictive model, they must expand the web scraping process to capture page counts and publisher details.

6. Conclusion
This project successfully scraped, cleaned, and analysed 1,000 books. While visualisations provided useful insights into price distributions, the linear regression model highlighted a key limitation: star ratings and genres are not sufficient to predict book prices. This demonstrates the value of data science in identifying which data points are truly valuable for business decisions.

7. References
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